CE/CS/SE 3354 · Software Engineering · Fall 2026

Project Milestone P2

Analysis and High-Level Design

Study Room Sniper – availability alerts for McDermott Library study rooms

Team	Team 4, section .003
Document version	1.0 (baseline for P2; grows from your P1 version 0.x)
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0 Revision History, Contribution Record and Changes Since P1

Version	Date	Author	Summary of change
0.1	16 October 2026	Team 4 (Alexander Eng, Arshan Khimani, Lee Baillard, Mayand Dhami, Henry McGahan, Isaam Memon)	P1 baseline
1.0	8 November 2026	Team 4	P2 baseline: analysis and high-level design added

0.1 Individual contribution record (P2)

Team member	Sections and artifacts this person produced for P2	Evidence in the repository (commit hashes, PR numbers)	Reviews given (PR numbers)
Alexander Eng	Change-control lead: document assembly, revision history, baseline and change tracking (§0, §1–§3 updates, §10.1, §12)	Final P2 PDF and .docx source in /docs/p2; commits for revision history, baseline and change tracking (sections 0, 1 to 3, 10.1, 12); git tag p2 on the commit containing the PDF.	(to be filled from the repository)
Arshan Khimani	Technical lead: Architecture Spike S-01 (§8), architecture draft and high-level design (§6, ADR-01 in §7), final P2 submission	Spike S-01 in /spikes/S-01 (script, request log, README with results); commits for the architecture draft and high-level design (section 6) and ADR-01 (section 7); Canvas submission.	(to be filled from the repository)
Lee Baillard	Elicitation lead: resolving open questions Q1–Q4 and Q6 (§11)	/docs/elicitation: messages sent to the library and OIT and any replies for Q1 to Q4 and Q6, with dates.	(to be filled from the repository)
Mayand Dhami	Quality lead: quality-attribute scenarios (§6.1), student survey analysis for Q5, accessibility verification (§6.6, §11)	Commits for the quality-attribute scenarios (6.1); Q5 student survey questions and analysis; accessibility verification notes (WCAG checklist or scan result) for 6.6 and QR-04.	(to be filled from the repository)

Team member	Sections and artifacts this person produced for P2	Evidence in the repository (commit hashes, PR numbers)	Reviews given (PR numbers)
Henry McGahan	Use case and modeling lead: expanded use cases (§4), domain model (§5.1), object interaction and sequence diagrams (§5.2)	/docs/p2/diagrams: sources and exported images for the use-case diagram, domain model and UC-01 and UC-02 sequence diagrams; commits for the expanded use cases (section 4) and domain model (5.1).	(to be filled from the repository)
Isaam Memon	Risk and integration lead: risk register (§10.3), document integration, consistency check (§5.3), traceability matrix extension (§9)	Commits for the risk register (10.3), consistency check (5.3) and traceability matrix (section 9); written notes on free-tier limits supporting R6.	(to be filled from the repository)

0.2 Changes since the P1 baseline

Change ID	What changed (IDs affected)	Why (P1 feedback, new finding, spike result)	Knock-on effect	Agreed on
CH-01	Sections 4 to 9 added: use-case analysis, domain and interaction models, high-level design, design rationale, architecture spike, extended traceability. No requirement ID (FR-01 to FR-14, QR-01 to QR-06, CON-01 to CON-03) was added, changed or withdrawn.	P2 milestone scope; spike S-01 outcome is pending (section 8).	Sections 1 to 3 unchanged; traceability extended in 9.1.	8 November 2026
CH-02	Plan to P3 filled in (10.2) and risk R6 added (10.3).	The hosting choice in ADR-04 depends on free-tier limits that could break QR-02.	None on requirements; R6 tracked by Isaam Memon.	8 November 2026

1 to 3 Introduction, Overall Description and Requirements (carried forward)

1 Introduction

1.1 Purpose of this document and of the system

This document is the milestone baseline for Team 4's CS/SE 3354 project. It states the problem that Study Room Sniper addresses, the scope the team has committed to, the requirements the system must meet, and the high-level design and analysis models. It is written for the course instructor and TA, who will assess it, and for the six team members, who will build from it. The requirements follow the style of ISO/IEC/IEEE 29148.

Study Room Sniper addresses a problem for students who need a study room at Eugene McDermott Library. Rooms are reserved through the library's online system, and when every room for the time a student needs is taken, the only way to learn that one has opened (a reservation is cancelled or lapses after the 15-minute hold) is to keep refreshing the booking page. Study Room Sniper lets a currently enrolled UTD student describe the room they need and alerts them as soon as a matching room becomes bookable. It does not book the room for them.

1.2 Scope

In scope: we will build this	Out of scope: we will not, and why
Watches for the 9 group study rooms and 8 smart rooms listed for McDermott Library	Booking, extending or cancelling any room on a student's behalf. It would need the student's library credentials or an agreement with the library, and could let one student crowd out others.
Read-only, periodic checking of room availability in the library's reservation system	Rooms outside McDermott Library, and the One Button Studio. Their booking rules are assumed to differ, and the team cannot verify them before P2.
Alerts by email and by browser push notification, each with a link to the library's booking page	SMS alerts. Messaging has a per-message cost, and the team has no budget (CON-03).
Sign-in with UTD NetID; each student sees and manages only their own watches	Native mobile apps. A responsive web interface reaches the same phones at far lower cost.
A basic operator view showing whether the checking of library availability is healthy	Predicting when rooms are likely to free up or recommending rooms. It needs historical data that the team does not have.

Why the exclusions were excluded. Booking on a student's behalf was excluded first and most firmly: it depends on permissions the team does not have and on credentials we do not want to hold, and an alerting tool is useful without it. The other exclusions were left out because each adds cost, data or verification effort that the six-person team cannot absorb, and none is needed for a student to learn that a room has opened.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
Study Room Sniper	The system specified in this document. Despite the name, it only alerts; it never reserves a room (CON-02).
Watch	A student's saved request to be alerted: a date, a time window and a minimum group size. Its status is Active, Expired, or Cancelled.
Slot	A period during which one study room can be reserved, as the library reservation system presents it.
Bookable	Two meanings, both in use: (1) Free: the library system shows the slot as unreserved. (2) Reservable by this student now: free, and also within the 7-day booking horizon and the student's own limits (one active reservation per 24 hours, at most four hours). Resolved in Q4; until then "bookable" in this document means free.
Match	A bookable slot that satisfies an Active watch: same date, entirely inside the watch's time window, in a room whose listed capacity fits the group size.
Read cycle (polling)	One pass in which the system reads current availability for every in-scope room from the library reservation system. Reads are read-only requests.
Alert	A message (email, optionally browser push) telling a student that a match exists.
Library Reservation System (LRS)	The library's online study-room booking service (UTD LibCal). External: the team can read from it but does not build or control it.
NetID; SSO	NetID is a student's UTD login identifier. SSO (single sign-on) lets a student sign in with it without creating a separate account.
Operator	A team member responsible for running the service and checking that it is healthy.
MoSCoW	Priority scheme used in section 3: MUST, SHOULD, COULD, WON'T.
FR, QR, CON	Identifier prefixes for functional, quality, and constraint requirements.
WCAG	Web Content Accessibility Guidelines, the accessibility standard cited in [R5].

1.4 References

- [R1] ISO/IEC/IEEE 29148:2018. Systems and Software Engineering — Life cycle processes — Requirements engineering.
- [R2] Eugene McDermott Library, "Study Rooms" (UTD LibCal), https://libcal.utdallas.edu/ — room counts, hours, 7-day booking horizon. Accessed 16 October 2026.
- [R3] Eugene McDermott Library, "Reserve a Study Room," https://library.utdallas.edu/about-the-library/reserve-a-study-room — enrolled students only; four-hour maximum; one active reservation per 24 hours; 15-minute hold. Accessed 16 October 2026.
- [R4] Eugene McDermott Library, "Space Availability - Study Rooms" policies, https://libcal.utdallas.edu/reserve/spaces/study-rooms — occupancy limits, floors and closing times. Accessed 16 October 2026.

- [R5] W3C, Web Content Accessibility Guidelines (WCAG) 2.2, https://www.w3.org/TR/WCAG22/.
- [R6] CS/SE 3354 Fall 2026 syllabus and Project Deliverables (course documents on Canvas).

2 Overall Description

2.1 Product perspective and context

Everything inside the system boundary is built and run by the team: the watch list, the periodic reading of availability, the matching of availability against watches, the sending of alerts and the operator health view. Everything the system calls on but does not build sits outside it: the library reservation system (the source of truth for availability, which we can read but not change), UTD NetID sign-on (which identifies students), and the email and push services that carry alerts. The boundary stops short of booking on purpose. The student, not the system, makes the reservation on the library's own site, so the system never holds library credentials and the library keeps control of who gets a room (CON-02).

2.2 Stakeholders

Stakeholder	Role in relation to the system	What they need from it	How we learned this
Students who want a group study room	Primary users. Create watches and receive alerts.	To learn quickly when a room matching their need opens, with little effort to set up, and an alert that arrives in time to act on.	Assumption, based on the team's position as UTD students and on the booking rules in [R2] and [R3]. Survey planned for Q5.
Students who use assistive technology or have other access needs	Primary users with specific constraints.	A watch list and alerts that work by keyboard and screen reader, with nothing conveyed by color or sound alone.	Assumption, taken from WCAG 2.2 [R5].
McDermott Library reservations staff	Operate the reservation system we read from. Never use ours.	That our traffic does not degrade their service, and that their rules for fair access to rooms (e.g. one active reservation per 24 hours [R3]) are not undermined.	Assumption, inferred from [R2]–[R4].
UTD Office of Information Technology (OIT)	Operates NetID sign-on and campus IT policy. Never uses ours.	That only approved services use NetID sign-on and that student data is limited to what the service needs.	Assumption.
Project team (Alexander Eng, Arshan Khimani, Lee Baillard, Mayand Dhami, Henry McGahan, Isaam Memon)	Builders and operators.	Free hosting, a service that is easy to monitor, and requirements stable enough to build against.	Direct knowledge: the team's own needs and the deadlines in [R6].

2.3 User characteristics

- **Students:** Currently enrolled UTD students, the only people allowed to reserve study rooms. Assumed comfortable with web apps, hold a NetID, and read most messages on a phone. They arrive in a hurry: either planning a session within the 7-day booking window or already in the library. Alerts must be readable at a glance on a phone, and creating a watch must take well under two minutes (QR-03).
- **Students with access needs:** Use screen readers, keyboard-only navigation, magnification, or reduced motion. The system must work for them in the same interface rather than a separate version (QR-04). Alerts cannot rely on sound or color alone.
- **Operators:** Team members who run the service. Comfortable with logs and web consoles and need only a simple view of when library availability was last read successfully and how many recent reads failed.

2.4 Constraints

- **CON-01:** Sign-in uses UTD NetID single sign-on only. Source: assumed OIT requirement for campus services; to be confirmed (Q3).
- **CON-02:** The system only reads from the library reservation system and never creates, changes or cancels a reservation. Source: library reservation policy, which ties each reservation to one enrolled student [R3], and the team's decision not to hold student credentials.
- **CON-03:** The system is built and run using only services that cost the team nothing. Source: team decision; the team has no budget.

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	The library reservation system lets a program read room availability without a student's credentials, for example from the public availability pages on libcal.utdallas.edu [R2].	FR-03 and FR-04 cannot be met as written. The team would have to hold student credentials (forbidden by CON-02) or fall back to demonstrating against recorded sample data.
A2	The library accepts automated, read-only checking at a modest rate: no more than 20 requests per minute in total (QR-05).	The library may block the service. The team would have to lower the read rate, weakening QR-01, or stop.
A3	UTD lets a student-built service use NetID sign-on, and releases at least a NetID and an email address.	FR-02 cannot be met. The team would raise a change request to CON-01 and use email-verified accounts, losing the assurance that users are UTD students.
A4	The set of rooms stays small: 17 group study and smart rooms today [R2], and no more than 20 in future.	More rooms would strain the request budget in QR-05: at 1 request/room, 20 requests/min covers at most 40 rooms every two minutes, so FR-03 could not be met.
A5	Students see an email or push alert on their phone within a few minutes while they are looking for a room.	Alerts arrive on time but go unseen and the benefit shrinks. Revisit alert channels and the SMS exclusion.
A6	Each room's capacity can be obtained from the library's pages [R4].	Matching on group size (FR-01, FR-04) is impossible. Group-size input would be dropped through a change request.

2.6 Feasibility summary

- **Technical:** Likely feasible; confirmed only once the spike on A1 is completed. Periodic reading, matching, and message sending involve no novel algorithms.
- **Operational:** Feasible, subject to the library not objecting (Q2). Students continue reserving via the library system; Study Room Sniper only adds an alert step.
- **Schedule:** Feasible for baseline; tight schedule across milestones.
- **Economic/resource:** Feasible; no budget needed using free-tier services (CON-03).

3 Specific Requirements

3.1 Functional requirements

ID	Requirement (one "shall" sentence)	Priority	Source	Acceptance criterion
FR-01	The system shall let a signed-in student create a watch by giving a date within the next 7 days, a time window and a minimum group size.	MUST	Students (2.2); [R2]	A tester creates a watch with each input and sees it in the watch list with the values entered; a date more than 7 days ahead is refused with a message.
FR-02	The system shall require a student to sign in with UTD NetID before creating, viewing or cancelling a watch.	MUST	CON-01	A tester tries each of the three functions while signed out and is sent to NetID sign-in each time; after signing in the function is available.
FR-03	The system shall read the availability of every in-scope room from the library reservation system at least once every 2 minutes while the library is open.	MUST	Students (2.2); [R2]	Against a stub library system that is always reachable, a tester reviews the request log for a 60-minute period and finds no room with a gap above 120 seconds between reads.
FR-04	The system shall recognize a match when a bookable slot that satisfies an Active watch appears in the availability data.	MUST	Students (2.2)	Using the stub, a tester releases one slot that fits a watch and one that does not (wrong time); within one read cycle the first is a match and the second is not.
FR-05	The system shall confirm that a slot is still bookable immediately before sending an alert for it.	SHOULD	Students (2.2)	Using the stub, a tester makes a matching slot disappear between the read and the send; no alert is sent for it.
FR-06	The system shall email the student an alert covering every match found in a read cycle.	MUST	Students (2.2)	Using the stub, a tester releases a matching slot and finds exactly one alert in the test inbox for the student who owns the watch.

ID	Requirement (one "shall" sentence)	Priority	Source	Acceptance criterion
FR-07	The system shall include in each alert the room name, the slot's start and end times and a link to the library's booking page.	MUST	Students (2.2); [R2]	A tester opens a received alert, checks that all three items are present and that the link opens the library booking page.
FR-08	The system shall let a student choose browser push as an additional alert channel for a watch.	SHOULD	Students (2.2)	A tester enables push on one watch and not on another and releases a match for each; the first produces an email and a push message, the second only an email.
FR-09	The system shall let a student cancel an Active watch.	MUST	Students (2.2)	A tester cancels an Active watch, then releases a slot that would have matched; no alert is sent and the watch shows as Cancelled.
FR-10	The system shall show a student a list of their own watches with the status of each (Active, Expired or Cancelled).	MUST	Students (2.2)	Using two accounts, a tester creates, cancels and expires watches; each account sees only its own watches, each with the correct status.
FR-11	The system shall mark a watch Expired when the end of its time window has passed and stop evaluating it.	MUST	Students; Library staff (2.2)	Using a test clock, a tester moves time past a watch's window end; within 5 minutes the watch is Expired and a later matching slot produces no alert.
FR-12	The system shall send at most one alert per slot for each watch.	MUST	Students (2.2)	With a slot that stays free across 10 read cycles, a tester receives exactly one alert for it.
FR-13	The system shall allow a student no more than 5 Active watches at one time.	SHOULD	Library staff; Team (2.2)	A tester creates 5 watches and tries a sixth, which is refused with a message; after cancelling one, a new watch can be created.
FR-14	The system shall show an operator the time of the last successful read of library availability and the number of failed reads in the last hour.	COULD	Team (2.2)	A tester stops the stub for 10 minutes; the operator view shows a last-read time that stops advancing and a failure count above zero, and both recover after the stub restarts.

3.2 Quality requirements

ID	Requirement (threshold, condition, method)	Priority	Source	How it will be verified
QR-01	Ninety-five percent of alerts shall be delivered to the student's email service within 180 seconds of the slot becoming bookable in the library system, measured under a load of up to 200 Active watches.	MUST	Students (2.2)	Scripted releases against the stub: compare each release time with the delivery time in the logs over 100 releases; pass if at least 95 are within 180 seconds.
QR-02	The service shall be available at least 99% of the time during library opening hours, measured over a continuous 14-day trial.	SHOULD	Students; Team (2.2)	Automated check every 5 minutes; availability is successful checks divided by all checks during opening hours, and must be at least 99%.
QR-03	At least 90% of first-time users shall create a watch in under 90 seconds without help, measured on a phone.	MUST	Students (2.2)	Moderated test with at least 10 UTD students, timed from first screen to watch appearing in list; pass if at least 9 of 10 succeed.
QR-04	Every screen used in UC-01 and UC-03 shall meet WCAG 2.2 Level AA [R5], with zero violations rated serious or critical.	MUST	Students with access needs (2.2); [R5]	Automated accessibility scan of each screen, plus keyboard-only and screen-reader walk-throughs of UC-01 and UC-03 where all steps can be completed.
QR-05	The system shall send no more than 20 requests per minute in total to the library reservation system, however many watches are Active.	MUST	Library staff (2.2); A2	Count requests in the request log over a 24-hour run with 200 Active watches; pass if no minute exceeds 20.
QR-06	The system shall store no personal data about a student other than their NetID, email address and watches.	MUST	OIT; Students (2.2)	Review a full export of stored data for three test accounts and confirm nothing else about the student is present.

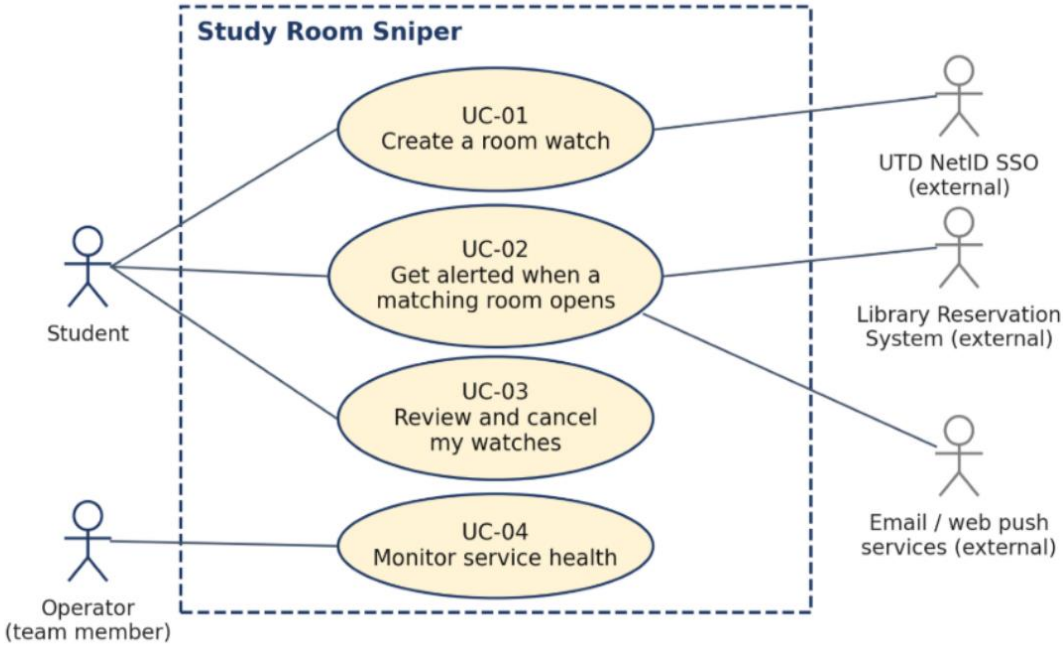
3.3 Constraint requirements

ID	Constraint (one sentence)	Source	Why it cannot be negotiated
CON-01	The system shall use UTD NetID single sign-on as its only means of student sign-in.	OIT (assumed, Q3)	The team has no other way to show that a user is a UTD student, and cannot issue accounts the library or OIT would trust.
CON-02	The system shall only read from the library reservation system and shall never create, change or cancel a reservation.	[R3]; team decision	Acting for a student needs credentials and library consent; neither exists, and holding credentials is an unacceptable security risk.

ID	Constraint (one sentence)	Source	Why it cannot be negotiated
CON-03	The system shall be built and run using only services that cost the team nothing.	Team (no budget)	The team has no funds, and a paid service would make the project impossible to keep running.

3.4 Use-case diagram and actor-goal summary

The use-case diagram is retained from P1, showing UC-01 to UC-04 with the Student, Operator, NetID SSO, Library Reservation System and Email / web push services. Step-by-step expansions are in section 4.



- **UC-01:** Create a room watch (Student, UTD NetID SSO)
- **UC-02:** Get alerted when a matching room opens (Student, Library Reservation System, Email / web push services)
- **UC-03:** Review and cancel my watches (Student, UTD NetID SSO)
- **UC-04:** Monitor service health (Operator)

3.5 Traceability

The requirements-to-architecture and implementation mapping is the extended traceability matrix in section 9.1.

4 Use-Case Analysis

4.1 Use cases selected for analysis

Use case	Name	Why it was selected	Non-trivial steps
UC-01	Create a room watch	Primary user-facing transaction carrying initial system state, authentication, input validation, and user quotas (FR-01, FR-02, FR-13).	Validation of window and horizon, checking the 5-watch cap, persisting the watch.
UC-02	Get alerted when a matching room opens	Carries core system value, involves two external systems (LRS, messaging), handles polling loops, matching logic, and drives performance QR-01.	Availability ingestion, match evaluation, de-duplication check, dispatching alerts.
UC-03	Review and cancel my watches	Manages watch lifecycle transitions (Active to Cancelled/Expired) and user privacy enforcement (FR-09, FR-10, FR-11).	Fetching the user-scoped watch list, transitioning status, updating the matcher cache.

4.2 Expanded use cases

UC-01 Create a room watch

Primary actor	Student
Supporting actors and external systems	UTD NetID SSO (identifies the student)
TUCBW (this use case begins when)	A student asks to be alerted about a study room.
TUCEW (this use case ends when)	The student sees the new watch in their watch list with status Active.
Preconditions	The student is a currently enrolled UTD student with a NetID.
Postconditions	A new watch owned by the student exists with status Active, and it appears in the student's watch list.
Traces to requirements	FR-01, FR-02, FR-10, FR-13; QR-03, QR-04, QR-06; CON-01

Actor input / action	System response
	(0) The system offers sign-in with UTD NetID.
(1) The student asks to sign in with NetID.	(2) The system identifies the student, recording them on first use, and shows the student's watch list with the status of each watch.
(3) The student enters the date, the time window and the minimum group size, chooses whether to also get browser push alerts, and confirms.	(4) The system checks that the date is within the next 7 days and the window starts before it ends, checks that the student has fewer than 5 Active watches, saves the watch as Active, and shows it in the list with status Active.

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	2	NetID sign-in does not complete (the student backs out or sign-on fails).	The system says the student could not be identified and offers to try again. No watch is created.	Ends (or restarts at 1)
A2	4	The date is in the past or more than 7 days ahead, or the window does not start before it ends.	The system refuses, says which entry is wrong and keeps the other entries.	Rejoins at 3
A3	4	The student already has 5 Active watches.	The system refuses and tells the student to cancel a watch first. No watch is created.	Ends; the student may cancel a watch (UC-03) and begin again at 3

UC-02 Get alerted when a matching room opens

Primary actor	Student
Supporting actors and external systems	Library Reservation System (reports free slots); Email / web push services (carry the alert)
TUCBW (this use case begins when)	A free slot that fits one of the student's Active watches appears in the library's availability.
TUCEW (this use case ends when)	The student opens the alert and sees the room name, the start and end times and the link to the library's booking page.
Preconditions	The student has an Active watch (created in UC-01) and has given an email address through NetID sign-in.
Postconditions	Exactly one alert for this slot and this watch is recorded and delivered to the student. No reservation has been made or changed.
Traces to requirements	FR-03, FR-04, FR-05, FR-06, FR-07, FR-08, FR-12; QR-01, QR-05; CON-02

Actor input / action	System response
	(0) The system shows the student's Active watch in the watch list and keeps reading the library's availability.
(1) The Library Reservation System shows a slot as free (a reservation was cancelled or lapsed).	(2) The system reads the free slot, finds that its date, times and room capacity fit the watch, finds no earlier alert for this slot and watch, and asks the library system to confirm the slot is still free.
(3) The Library Reservation System confirms the slot is still free.	(4) The system records the alert and sends it by email, and by browser push if the watch has push enabled.
(5) The student opens the alert.	(6) The alert shows the room name, the slot's start and end times and the link to the library's booking page.

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	3	The slot has been taken by the time the library system is asked.	The system sends no alert, records nothing for this slot and keeps the watch Active.	Ends
A2	2	The library's availability cannot be read in this cycle.	The system records the read as failed, sends no alerts from it and tries again in the next cycle.	Rejoins at 1
A3	2	An alert for this slot and this watch was already sent.	The system sends no second alert.	Ends

UC-03 Review and cancel my watches

Primary actor	Student
Supporting actors and external systems	UTD NetID SSO (identifies the student)
TUCBW (this use case begins when)	A student asks to see their watches.
TUCEW (this use case ends when)	The student sees the chosen watch with status Cancelled.
Preconditions	The student is a currently enrolled UTD student with a NetID.
Postconditions	The chosen watch has status Cancelled and is no longer evaluated; the student's other watches are unchanged.
Traces to requirements	FR-02, FR-09, FR-10, FR-11; QR-04, QR-06; CON-01

Actor input / action	System response
	(0) The system offers sign-in with UTD NetID.
(1) The student asks to sign in with NetID.	(2) The system identifies the student and shows only that student's watches, each with its status (Active, Expired or Cancelled).
(3) The student chooses an Active watch and asks to cancel it.	(4) The system asks the student to confirm cancelling that watch.
(5) The student confirms.	(6) The system marks the watch Cancelled, stops evaluating it, and shows the list with the watch as Cancelled.

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	2	The student has no watches.	The system says there are none and offers to create one (UC-01).	Ends
A2	2	NetID sign-in does not complete.	The system says the student could not be identified and offers to try again.	Ends (or restarts at 1)
A3	5	The student declines to confirm.	The system leaves the watch Active and shows the list again.	Rejoins at 2
A4	6	The watch became Expired while the student was deciding.	The system says the watch is already Expired and changes nothing.	Ends

4.3 Self-review log

Use case	Check from the Lesson 13 review checklist	Defect found	How it was fixed
UC-01	Every item the actor column names is an attribute in the domain model	None found: NetID, email, date, window start and end, group size, push choice and status are all attributes in 5.1.	No change needed.
UC-02	No background processing	The first draft started with the system reading availability, with no actor and nothing the student could see or wait for.	Made the Library Reservation System the actor of step (1) and added step (0) showing the student's Active watch; the student's wait ends at the alert in step (6).

Use case	Check from the Lesson 13 review checklist	Defect found	How it was fixed
UC-03	Precondition or alternate flow	"The student has at least one watch" was written as a precondition, but it can be false while the system works normally.	Moved to alternate flow A1 at step (2); the precondition now says only that the student has a NetID.

5 Analysis Models

5.1 Domain model

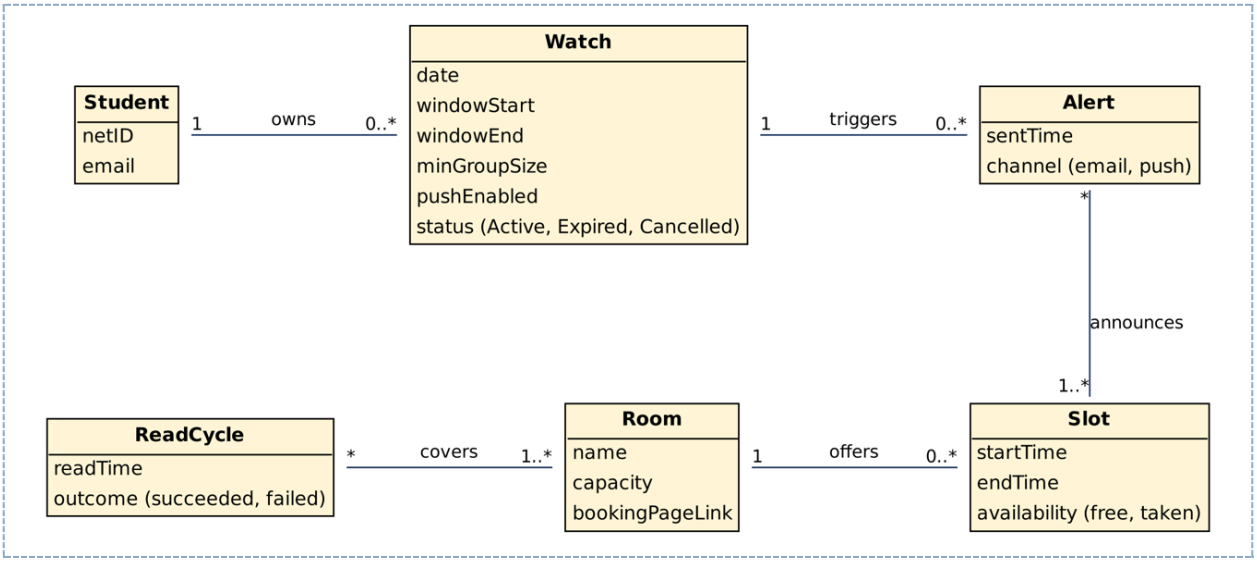


Figure 5.1 Domain model (conceptual classes, attributes, associations with multiplicities)

Class	What it represents (one line)	Key attributes	Associations (with multiplicity)
Student	A currently enrolled UTD student who uses the service.	netID, email	Owns 0..* Watch; each Watch has exactly 1 Student.
Watch	A student's saved request to be alerted about a room.	date, windowStart, windowEnd, minGroupSize, pushEnabled, status (Active, Expired, Cancelled)	Owned by 1 Student; triggers 0..* Alert; each Alert has exactly 1 Watch.
Alert	A message telling a student that a match exists.	sentTime, channel (email, push)	Triggered by 1 Watch; announces 1..* Slot; each Slot is announced by 0..* Alert.
Room	A study room (group study or smart room) at McDermott Library.	name, capacity, bookingPageLink	Offers 0..* Slot; each Slot belongs to 1 Room; covered by 0..* ReadCycle.
Slot	A period during which one room can be reserved.	startTime, endTime, availability (free, taken)	Offered by 1 Room; announced by 0..* Alert.
ReadCycle	One pass that reads availability for the in-scope rooms.	readTime, outcome (succeeded, failed)	Covers 1..* Room; each Room is covered by 0..* ReadCycle.

5.2 Object interaction models

5.2.1 UC-01 Create a room watch

(a) Non-trivial steps.

Three steps of UC-01 carry real logic: identifying the student through NetID sign-on (use-case steps 1 and 2, scenario rows 1 to 4); checking the date, the window order and the 5-watch limit (use-case step 4, rows 8 and 9); and creating the watch with status Active (use-case step 4, rows 10 and 11). WatchManager is an interaction object, not a domain class: it is the one object in control, standing for the part of the system that accepts requests about watches.

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
1	Student (actor)	asks to sign in with NetID	NetID	WatchManager
2	WatchManager	asks to identify the student	—	NetID SSO
3	NetID SSO	returns NetID and email	NetID, email	WatchManager
4	WatchManager	finds or records the student	NetID, email	Student
5	WatchManager	asks for the student's watches	—	Student
6	WatchManager	shows the watch list with statuses	watches, status	Student (actor)
7	Student (actor)	confirms date, window, group size, push choice	date, window, group size, push choice	WatchManager
8	WatchManager	checks date is within 7 days and window order	date, window	WatchManager
9	WatchManager	asks how many Active watches the student has	—	Student
10	WatchManager	creates the watch	date, window, group size, push choice	Watch
11	Watch	sets status Active	status	Watch
12	WatchManager	shows the saved watch, status Active	watch, status	Student (actor)

(c) Analysis sequence diagram.

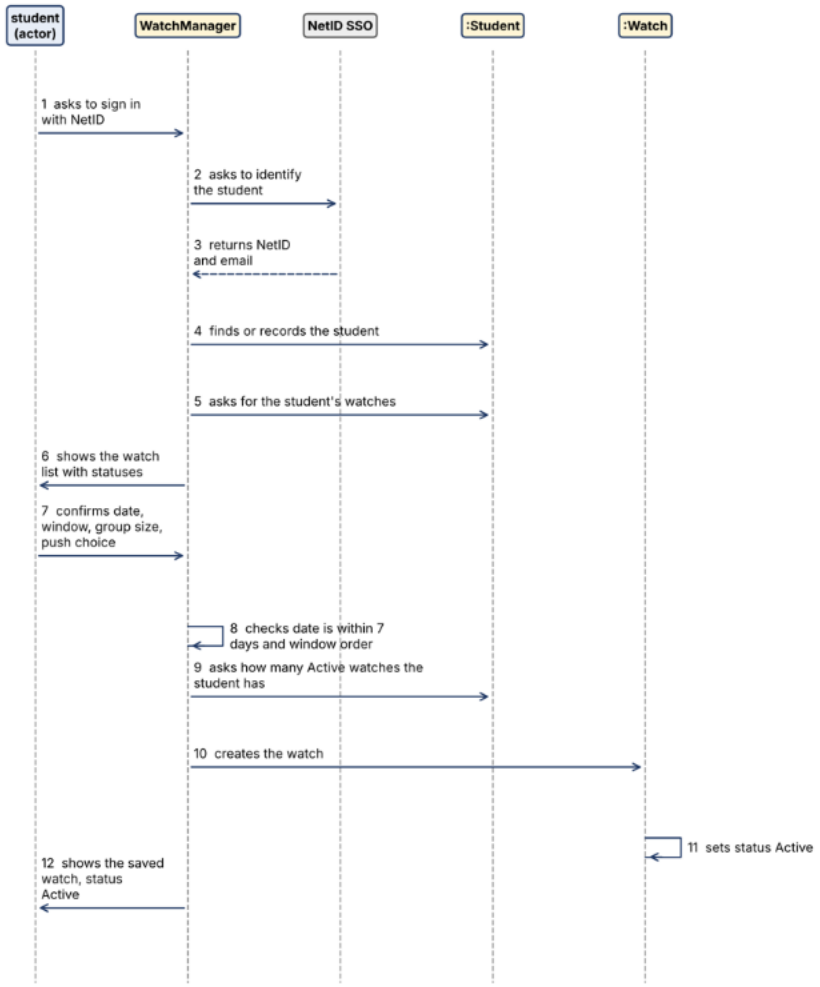


Figure 5.2.1 Analysis sequence diagram for UC-01; message numbers match the scenario table

5.2.2 UC-02 Get alerted when a matching room opens

(a) Non-trivial steps.

Three steps of UC-02 carry real logic: reading availability and recording the read (use-case step 2, scenario rows 1 to 3); matching free slots to Active watches and checking for an earlier alert (rows 4 to 6); and confirming the slot, creating the alert and delivering it (use-case steps 2 to 4, rows 7 to 10). AvailabilityMonitor is an interaction object, not a domain class: it is the one object in control, standing for the part of the system that drives each read cycle.

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
1	AvailabilityMonitor	asks for availability of every in-scope room	in-scope rooms	Library Reservation System

2	Library Reservation System	returns free slots per room	free slots, room	AvailabilityMonitor
3	AvailabilityMonitor	records the read (time, succeeded)	time, outcome	ReadCycle
4	AvailabilityMonitor	asks for Active watches	—	Watch
5	AvailabilityMonitor	asks whether a free slot fits a watch	free slot, date, window, group size	Watch
6	AvailabilityMonitor	asks whether the slot was already alerted	slot, watch	Alert
7	AvailabilityMonitor	asks to confirm the slot is still free	slot, room	Library Reservation System
8	AvailabilityMonitor	creates and records the alert	room, start/end time, booking link	Alert
9	AvailabilityMonitor	asks to deliver the alert	alert, student email	Email / Push Service
10	Email / Push Service	delivers email (and push) alert	alert	Student (actor)

(c) Analysis sequence diagram.

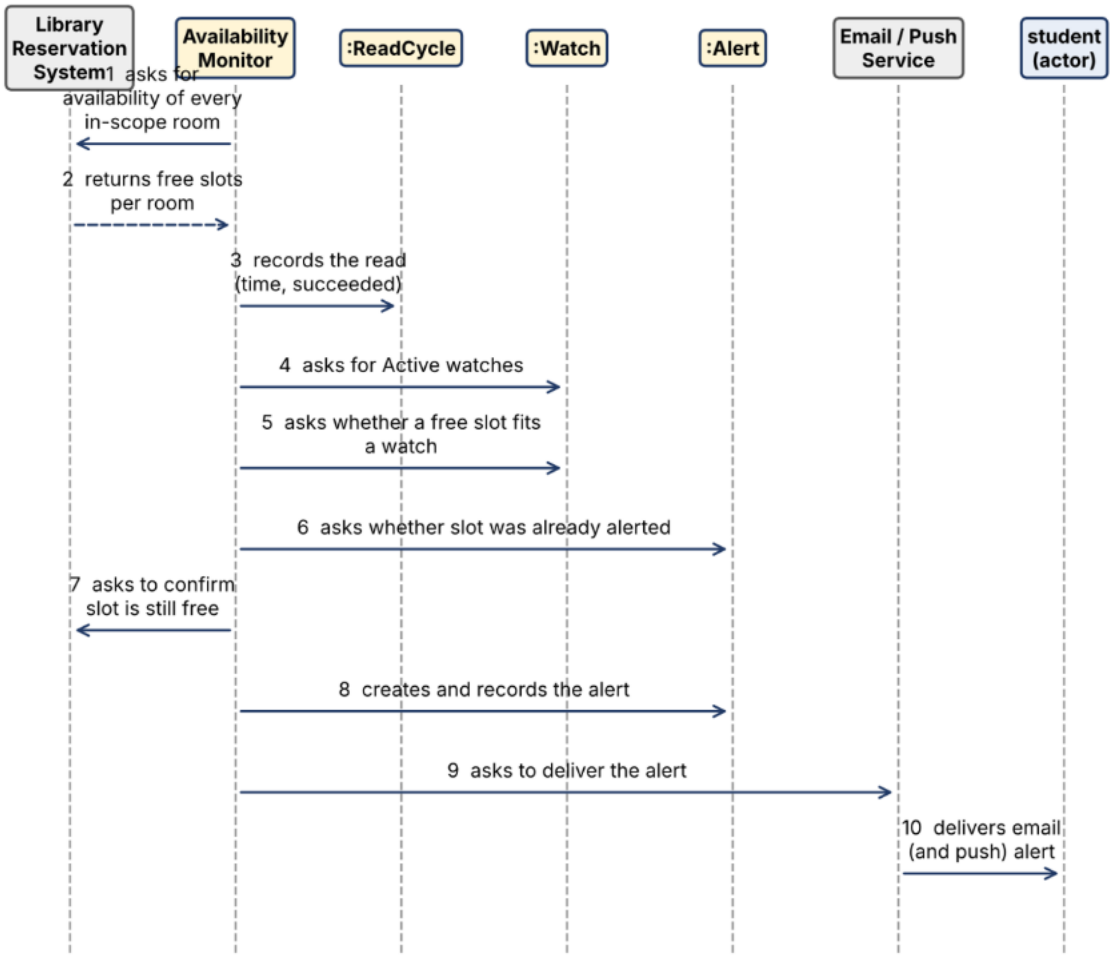


Figure 5.2.2 Analysis sequence diagram for UC-02; message numbers match the scenario table

5.3 Consistency cross-check

Check	Result (pass, or what failed)	Fixed in
K1 Every item in an actor column (4.2) is an attribute in the domain model (5.1)	Pass: all items appear as attributes of Student, Watch or Room.	n/a
K2 Every object on a sequence diagram is a domain class or a declared interaction object	Failed at first: WatchManager and AvailabilityMonitor are not in 5.1. Other objects are domain classes or external systems.	5.2 (a): both declared as interaction objects
K3 Every message on a sequence diagram corresponds to exactly one scenario-table row	Pass: 12 messages for UC-01 and 10 for UC-02, numbered as in the tables.	n/a
K4 Every non-trivial step in 5.2 (a) has at least one scenario-table row	Pass: each step named in (a) cites its rows.	n/a

Check	Result (pass, or what failed)	Fixed in
K5 Every analyzed use case traces to at least one functional requirement	Pass: UC-01 to FR-01, FR-02, FR-10, FR-13; UC-02 to FR-03 to FR-08 and FR-12; UC-03 to FR-02, FR-09 to FR-11.	n/a
K6 The last messages of each diagram make the use case postcondition true	Pass: UC-01 ends with the saved watch shown as Active; UC-02 ends with the alert recorded (row 8) and delivered (row 10).	n/a

6 High-Level Design (Architecture)

6.1 Architectural drivers: quality-attribute scenarios

QAS	Quality attribute	Stimulus	Artifact	Response	Response measure	From
QAS-01	Performanc e	A slot that fits an Active watch becomes free while 200 watches are Active.	Read, match and alert pipeline (Cycle Coordinator , Availability Reader, Match Engine, Alert Dispatcher).	The match is recognized and the alert is handed to the email service.	95% of alerts delivered within 180 s of the slot becoming bookable.	QR-01 (FR-03, FR-06)
QAS-02	Availability	The library system cannot be reached for 10 minutes during opening hours.	Availability Reader, Cycle Coordinator , Health View.	Failed reads are recorded and retried each cycle with no false alerts; reading resumes by itself.	At least 99% of 5-minute checks succeed over 14 days; first good read within 120 s of recovery.	QR-02 (FR-14)
QAS-03	Efficiency (external load)	Active watches grow from 10 to 200.	Availability Reader.	One shared read per room per cycle, whatever the number of watches.	No minute in a 24-hour run exceeds 20 requests to the library system.	QR-05 (CON-02)
QAS-04	Security and privacy	A signed-out visitor asks for watches, or a student asks for another student's watches.	Watch Service, Identity Adapter, Watch Store.	The visitor is sent to NetID sign-in; a student sees only their own watches; only NetID, email and watches are stored.	3 of 3 signed-out functions redirect; a data export for 3 test accounts holds nothing else.	QR-06 (FR-02, FR-10)
QAS-05	Usability	A first-time student on a phone creates a watch.	Web Client, Watch Service.	The watch is saved and shown as Active.	9 of 10 students finish in under 90 s unaided; zero serious or critical WCAG 2.2 AA violations on UC-01 and UC-03 screens.	QR-03, QR-04

Constraints that shape the architecture: CON-01 fixes sign-in through UTD NetID single sign-on, an external identity provider; CON-02 means no component may write to the library system; CON-03 limits hosting and messaging to free services, which rules out paid messaging, brokers and extra servers.

6.2 Architectural style and alternatives

Style considered	Drivers it serves	What it costs	Verdict
Layered client-server, deployed as one application with a read, match, alert pipeline inside it	QAS-01, QAS-03, QAS-04, QAS-05; CON-03	Components cannot scale or fail separately; one fault can stop the whole back end.	Chosen
Microservices (a separate service for watches, reading, matching and alerting)	Independent scaling and fault isolation (QAS-02).	Several deployments, network hops and monitoring for six students on free tiers (CON-03); more places for QAS-01 delay.	Rejected
Event-driven, with a message broker between reading, matching and alerting	Loose coupling between reading and alerting.	A broker the team must host free of charge (CON-03), extra failure modes and harder testing, for at most 17 rooms and 200 watches.	Rejected

The team chose the layered client-server style because the scenarios ask for one place that reads each room once per cycle and passes the result straight to matching and alerting (QAS-01, QAS-03), and for one back end that owns identity and data behind a thin web client (QAS-04, QAS-05). Microservices and an event broker would give independent scaling that 17 rooms and 200 watches do not need, at a running cost the team cannot pay (CON-03). ADR-01 records the decision.

6.3 Component view

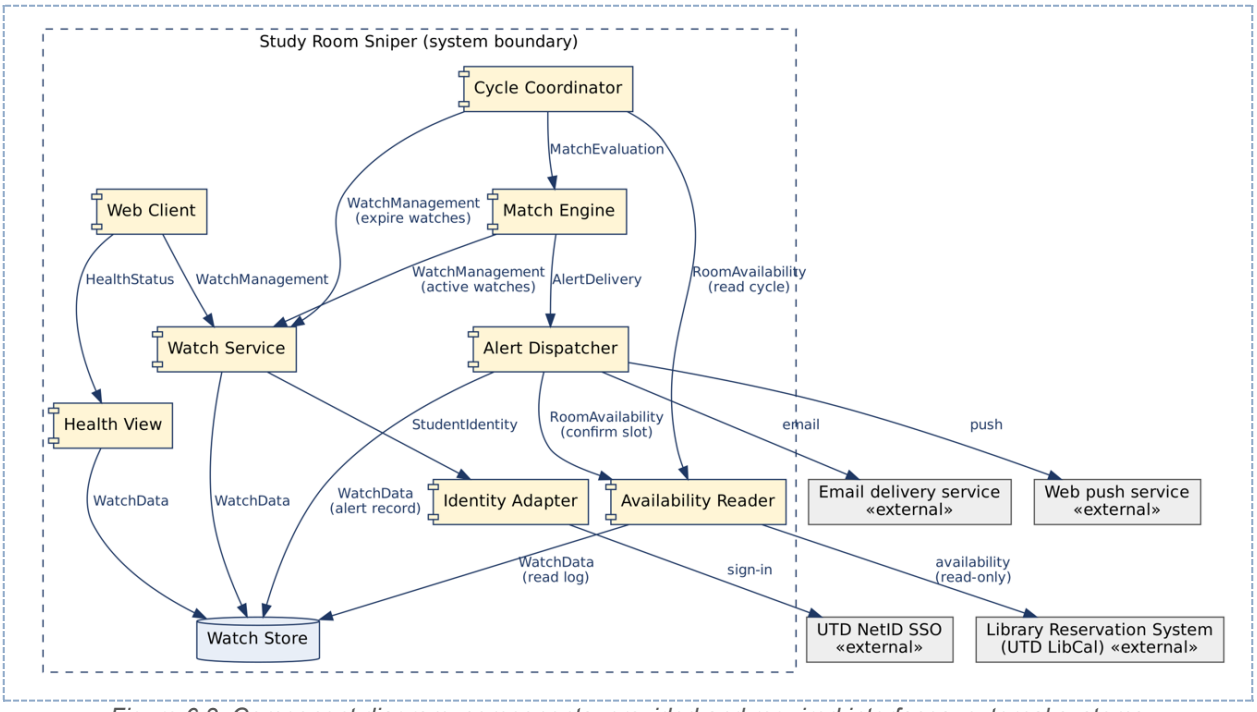


Figure 6.3 Component diagram: components, provided and required interfaces, external systems

Component	Responsibility (one sentence)	Provides	Requires	Answers to (UC / QAS / ADR)
Web Client	Shows the watch list, the watch form and the operator health page in the student's browser.	(none; user interface)	WatchManagement, HealthStatus	UC-01, UC-03, UC-04; QAS-05
Watch Service	Creates, lists, cancels and expires watches and enforces the 5-watch limit.	WatchManagement	StudentIdentity, WatchData	UC-01, UC-03; QAS-04
Identity Adapter	Signs students in through NetID single sign-on and passes on NetID and email.	StudentIdentity	NetID SSO sign-in (external)	UC-01, UC-03; QAS-04; ADR-03; CON-01
Cycle Coordinator	Starts each read cycle, hands results to matching and expires finished watches.	(none; drives the cycle)	RoomAvailability, MatchEvaluation, WatchManagement	UC-02; QAS-01, QAS-03; ADR-01
Availability Reader	Reads room availability from the library system, read-only, within the request budget, and logs each read.	RoomAvailability	Library system availability (external), WatchData	UC-02; QAS-02, QAS-03; ADR-02; CON-02
Match Engine	Decides which free slots satisfy which Active watches.	MatchEvaluation	WatchManagement, AlertDelivery	UC-02; QAS-01
Alert Dispatcher	Checks no earlier alert exists, confirms the slot, records the alert and sends it by email and push.	AlertDelivery	RoomAvailability, WatchData, email and push services (external)	UC-02; QAS-01
Health View	Shows the last successful read and the failed reads in the last hour.	HealthStatus	WatchData	UC-04; QAS-02
Watch Store	Holds students, watches, alerts and the read log.	WatchData	(none)	All use cases; QAS-04; ADR-04

6.4 Deployment view

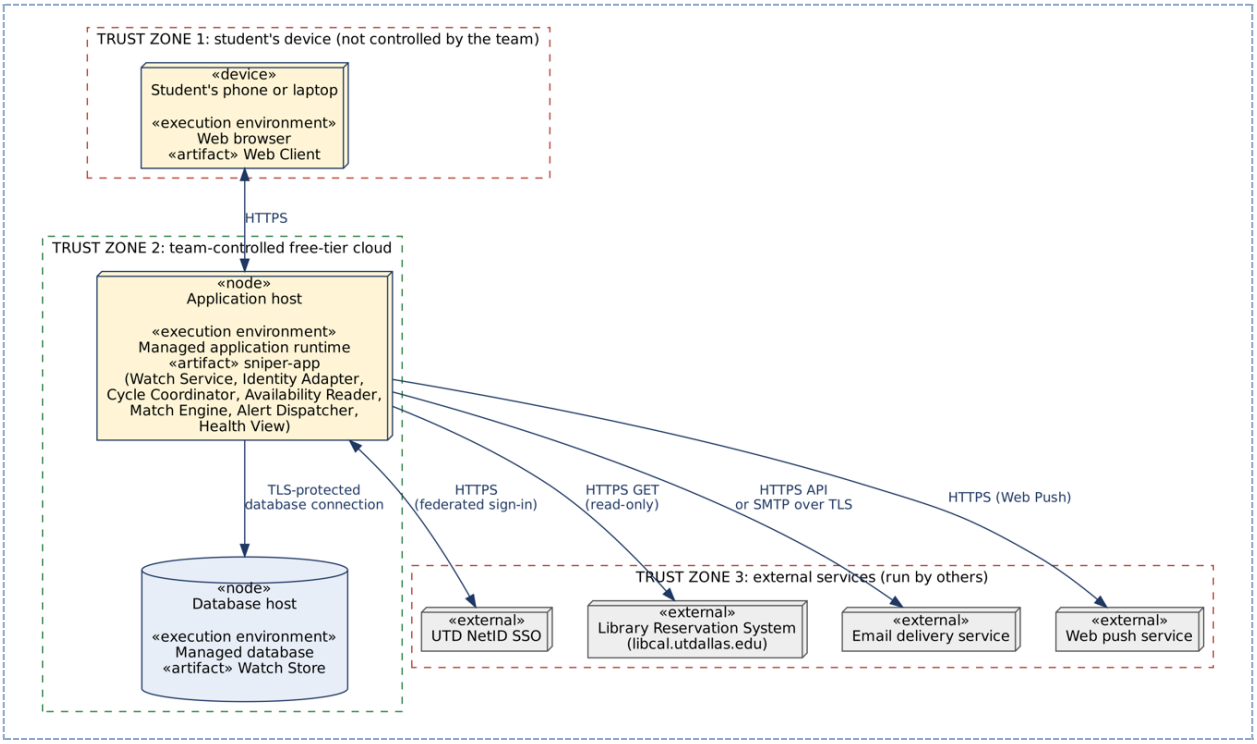


Figure 6.4 UML deployment diagram: nodes, execution environments, deployed artifacts, communication paths with protocols, trust boundaries

Node	What runs on it (artifacts, components)	Communicates with (protocol)	Trust zone
Student device	Web browser running the Web Client.	Application host over HTTPS.	Zone 1: student device, not controlled by the team
Application host	Managed application runtime running sniper-app: Watch Service, Identity Adapter, Cycle Coordinator, Availability Reader, Match Engine, Alert Dispatcher, Health View.	Student device (HTTPS); Database host (TLS-protected database connection); NetID SSO (HTTPS, federated sign-in); library system (HTTPS GET, read-only); email service (HTTPS API or SMTP over TLS); web push service (HTTPS, Web Push).	Zone 2: team-controlled free-tier cloud
Database host	Managed database holding the Watch Store.	Application host (TLS-protected database connection).	Zone 2: team-controlled free-tier cloud
External services	UTD NetID SSO, Library Reservation System (libcal.utdallas.edu), email delivery service, web push service.	Application host, using the protocols listed above.	Zone 3: external services run by others

Trust boundaries lie between zones 1 and 2 and between zones 2 and 3. Data crossing them: browser input arrives from an uncontrolled device and is validated; NetID and email arrive from the sign-on service; read-only requests with no credentials go out to the library system; alerts carrying the student's email, room and times go out to the email and push services. Under the shared-responsibility model the cloud providers secure the physical infrastructure, the runtime platform and the database engine, while the team remains responsible for the application code, its configuration, secrets (kept out of the repository), access control, the data stored and the TLS settings.

6.5 From analysis to architecture

Analysis object (from 5.1 and 5.2)	Component that owns it (6.3)	Note (split, merged, external)
Student (class)	Watch Service	Record kept in the Watch Store; identity comes from the NetID SSO via the Identity Adapter.
Watch (class)	Watch Service	Lifecycle owned here; stored in the Watch Store.
Alert (class)	Alert Dispatcher	Created and sent here; record kept in the Watch Store.
Room and Slot (classes)	Availability Reader	Source of truth is the library system (external); the reader only reads it.
ReadCycle (class)	Availability Reader	Recorded here; scheduled by the Cycle Coordinator and shown by the Health View.
WatchManager (interaction object, UC-01)	Watch Service	Sign-in steps split off to the Identity Adapter; the Web Client is the student-facing side.
AvailabilityMonitor (interaction object, UC-02)	Cycle Coordinator	Split: cycle control (Cycle Coordinator), reading (Availability Reader), matching (Match Engine), confirming and delivering (Alert Dispatcher).
NetID SSO, Library Reservation System, Email / Push Service	External systems	Outside the system boundary; reached through the adapters above.

6.6 Design principles applied

Principle	Where it shows in your design	Evidence (component, interface, ADR)
Separation of concerns	Each concern has its own component: sign-in, watch management, reading, matching, alerting, operator view.	Figure 6.3; components in 6.3; ADR-01

Principle	Where it shows in your design	Evidence (component, interface, ADR)
Information hiding	Components talk only through named interfaces; the library's page format and the NetID protocol are known only to Availability Reader and Identity Adapter.	Provided and required interfaces in 6.3; ADR-02, ADR-03
Low coupling	Components depend on interfaces such as WatchManagement and RoomAvailability, not on one another's internals; only the Watch Store touches the database.	Figure 6.3 (WatchData is the only path to the store)
High cohesion	Each component has one job stated in one sentence; the Alert Dispatcher both confirms and sends because the two steps must happen together (FR-05).	Responsibility column in 6.3
Design for change	The parts most likely to change, the library source and the alert channels, sit behind Availability Reader and Alert Dispatcher; a new channel touches one component.	ADR-02; QAS-03; Alert Dispatcher in 6.3
Keep it simple (KISS)	One application, one database and a fixed read cycle; no broker, no separate services, no booking.	ADR-01; style table in 6.2; CON-02

7 Design Rationale

7.1 Architecture decision records

ADR-01 Layered client-server application as one deployable unit

Status	Accepted
Date and owner	8 November 2026; Arshan Khimani
Context (drivers)	QAS-01 (alert delay), QAS-03 (request budget) and QAS-05 need a simple, fast path from reading to alerting; CON-03 and a six-person team limit what can be run.
Options considered	(a) Layered client-server, one application. (b) Microservices. (c) Event-driven with a message broker.
Decision	Option (a): one application with the components of 6.3, a thin web client and one managed database.
Consequences	Favors simplicity, testability, low cost and a short alert path. We accept that components cannot scale or fail separately and that a fault in one can stop the whole back end.
Evidence	None yet; spike S-01 (section 8) measures a full read cycle inside one application. Open question: Q1.
Related	QAS-01, QAS-03, ADR-04, R1, R4

ADR-02 One shared reader with a single request budget

Status	Accepted
Date and owner	8 November 2026; Arshan Khimani
Context (drivers)	QAS-01 and QAS-03; QR-05 (20 requests per minute); A1, A2 and A4; CON-02.
Options considered	(a) One shared read per room per cycle, whatever the number of watches. (b) One read per watch. (c) Read on demand when a student opens the page.
Decision	Option (a): the Availability Reader reads each in-scope room once per cycle, at most every 2 minutes, which is 17 requests per 2 minutes (8.5 per minute) against the cap of 20.
Consequences	Favors a load on the library that does not grow with watches (QR-05) and a steady alert delay (QR-01). We accept that freshness is limited to the read interval and that one shared pipeline is a single point of failure.
Evidence	None yet; spike S-01 checks that one request per room returns its slots and capacity. Open questions: Q1, Q2.
Related	QAS-01, QAS-03, QR-05, R1, R2

ADR-03 Delegate sign-in to UTD NetID single sign-on

Status	Accepted
Date and owner	8 November 2026; Lee Baillard
Context (drivers)	QAS-04; CON-01; QR-06; FR-02; assumption A3.
Options considered	(a) UTD NetID single sign-on. (b) Our own accounts with email verification. (c) Asking for library credentials.
Decision	Option (a): sign-in goes through NetID single sign-on, and the system stores only NetID, email and watches.
Consequences	Favors security and privacy (QAS-04, QR-06) and the assurance that users are UTD students, with no passwords held. We accept that the service depends on OIT allowing it (Q3, R3) and that nobody can sign in when single sign-on is down.
Evidence	None yet; Q3 is a pending inquiry with OIT.
Related	QAS-04, CON-01, QR-06, R3, Q3

ADR-04 Managed free-tier hosting and database

Status	Accepted
Date and owner	8 November 2026; Isaam Memon
Context (drivers)	CON-03 (no budget); QAS-02 (availability); a six-person team that cannot run its own servers.
Options considered	(a) Managed free-tier application runtime and managed database. (b) Self-hosting on a team member's machine. (c) A paid host.
Decision	Option (a): the application and database run on managed free tiers (Figure 6.4).
Consequences	Favors zero cost and little operating work (CON-03). We accept free-tier quotas and possible idle behavior that could threaten QR-02, and some dependence on the provider.
Evidence	None yet; open item for P3: check the chosen provider's idle and quota rules against a 2-minute read cycle (R6).
Related	QAS-02, CON-03, ADR-01, R6

7.2 Trade-off summary

Quality favored	Architectural move	Cost accepted	Recorded in
Simplicity, low cost, short alert path	One application with one database (ADR-01)	Components cannot scale or fail separately.	ADR-01
Steady load on the library (QR-05) and alert delay (QR-01)	One shared reader with a single request budget (ADR-02)	Freshness limited to the read interval; one pipeline is a single point of failure.	ADR-02

Quality favored	Architectural move	Cost accepted	Recorded in
Security and privacy (QR-06)	Sign-in delegated to NetID single sign-on (ADR-03)	Depends on OIT approval and on single sign-on being available.	ADR-03
Cost (CON-03)	Managed free-tier hosting and database (ADR-04)	Free-tier quotas and idle behavior may threaten availability (QR-02).	ADR-04

8 Prototype Evidence: Architecture Spike

Spike ID and title	S-01 Read library availability without signing in
Risk or decision it addresses	R1 (library availability cannot be read by a program); A1, A2; ADR-02; QAS-01, QAS-03; Q1
Question (answerable yes or no)	Can a program read the free and taken slots and the capacity of all 17 McDermott group study and smart rooms from the library's public availability pages without signing in, using one request per room per read cycle, and finish a full cycle in under 60 seconds?
Set-up	A small script in /spikes/S-01 reads each room once per cycle, paced at no more than 20 requests per minute, for 60 minutes within library opening hours. No credentials are used or stored. The test stub from section 3 is not involved.
How it was measured	The script logs, per request: time, room, HTTP status and whether slots and capacity were parsed. From the log: success rate per room, duration of each full cycle, highest requests in any minute, and the number of free-slot changes seen.
Result (numbers)	[TO BE FILLED by Arshan Khimani from the committed log: rooms parsed out of 17; longest cycle in seconds; highest requests per minute; failed requests.]
What it changed or confirmed	If yes: confirms A1 and ADR-02 and closes Q1 and R1. If no: a change request to FR-03 and FR-04 is raised and the team falls back to recorded sample data, as R1 states.
Where to see it	/spikes/S-01/ with README and request log; commit hash [TO BE FILLED].

9 Traceability

9.1 Extended traceability matrix

Requirement	Use case	Analysis classes	Component(s)	QAS / ADR	Evidence
FR-01	UC-01	Student, Watch	Web Client, Watch Service, Watch Store	QAS-05	5.2.1 rows 7 to 12
FR-02	UC-01, UC-03	Student	Identity Adapter, Watch Service	QAS-04; ADR-03	5.2.1 rows 1 to 4
FR-03	UC-02	ReadCycle, Room, Slot	Cycle Coordinator, Availability Reader	QAS-01, QAS-03; ADR-02	5.2.2 rows 1 to 3; spike S-01 (pending)
FR-04	UC-02	Watch, Slot, Room	Match Engine	QAS-01; ADR-01	5.2.2 rows 4 and 5
FR-05	UC-02	Slot, Room	Alert Dispatcher, Availability Reader	QAS-01	5.2.2 row 7
FR-06	UC-02	Alert, Watch, Student	Alert Dispatcher	QAS-01	5.2.2 rows 8 to 10
FR-07	UC-02	Alert, Room, Slot	Alert Dispatcher	QAS-01	5.2.2 rows 8 to 10
FR-08	UC-01, UC-02	Watch, Alert	Web Client, Watch Service, Alert Dispatcher	ADR-01	5.2.1 row 7; 5.2.2 row 9
FR-09	UC-03	Watch	Web Client, Watch Service	ADR-01	4.2 UC-03 steps 3 to 6
FR-10	UC-01, UC-03	Student, Watch	Watch Service, Watch Store	QAS-04	5.2.1 rows 5 and 6; 4.2 UC-03 step 2
FR-11	UC-03	Watch	Cycle Coordinator, Watch Service	ADR-01	4.2 UC-03 step 2
FR-12	UC-02	Alert, Watch, Slot	Match Engine, Alert Dispatcher, Watch Store	QAS-01	5.2.2 row 6
FR-13	UC-01	Student, Watch	Watch Service	ADR-01	5.2.1 row 9
FR-14	UC-04 (not analyzed)	ReadCycle	Health View, Availability Reader	QAS-02	None yet
QR-01	UC-02	Alert, Slot, Watch	Cycle Coordinator, Availability Reader, Match Engine, Alert Dispatcher	QAS-01; ADR-01, ADR-02	Spike S-01 (pending)

Requirement	Use case	Analysis classes	Component(s)	QAS / ADR	Evidence
QR-02	UC-02, UC-04	ReadCycle	Cycle Coordinator, Availability Reader, Health View	QAS-02; ADR-04	None yet
QR-03	UC-01	Watch	Web Client	QAS-05	None yet (usability test planned)
QR-04	UC-01, UC-03	(screens only)	Web Client	QAS-05	None yet (Q6 pending)
QR-05	UC-02	ReadCycle, Room	Availability Reader, Cycle Coordinator	QAS-03; ADR-02	Spike S-01 (pending)
QR-06	UC-01	Student, Watch, Alert	Watch Store, Watch Service, Identity Adapter	QAS-04; ADR-03	Deployment view 6.4
CON-01	UC-01, UC-03	Student	Identity Adapter	ADR-03; QAS-04	Q3 pending
CON-02	UC-02	Slot, Room	Availability Reader (read-only)	ADR-02	Deployment view 6.4 (HTTPS GET only)
CON-03	(all)	(none)	Application host, Database host	ADR-04, ADR-01	None yet

9.2 Gap analysis

Gap	Items	What you did about it
Requirements with no component	None.	Every requirement in 9.1 reaches at least one component; FR-14 is realized by the Health View and CON-03 by the two hosts.
Components with no requirement, QAS or ADR	None.	Every component in 6.3 answers to at least one use case, scenario or decision.
Use cases not analyzed (and why that is acceptable)	UC-04 Monitor service health	COULD priority (FR-14), one actor, one screen that reads data already analyzed (ReadCycle). It will be expanded in P3 if FR-14 is kept.

10 Project Plan (updated; was P1 section 4)

10.1 Roles and working agreement

Unchanged since v0.1. No roles have moved between members.

- **Alexander Eng:** Document and change-control lead.
- **Arshan Khimani:** Technical lead (spike, feasibility, repo tooling, architecture).
- **Lee Baillard:** Elicitation lead (library and OIT communications).
- **Mayand Dhami:** Quality and accessibility lead (WCAG checks, survey, usability tests).
- **Henry McGahan:** Use case lead (use case models, domain model, interaction sequences).
- **Isaam Memon:** Risk and integration lead (risk register, free-tier limits, document integration).
- **Decision rule:** Each owner decides within their area after hearing the others, with evidence recorded in the issue. If an objection is raised, a simple majority of the six decides. A tie or deadlock of 48 hours escalates to the instructor or TA.

10.2 Plan to P3

Milestone	Date	Owner	Done when
Spike S-01 result recorded; ADR evidence updated	13 November 2026	Arshan Khimani	Log and README committed in /spikes/S-01; section 8 and ADR-01 and ADR-02 evidence updated.
Replies on Q2, Q3 and Q6 received; CON-01 confirmed or change request raised	20 November 2026	Lee Baillard	Each question marked answered in section 11, or a change request recorded in 0.2.
Survey (Q5) analyzed; usability and accessibility test plan written	20 November 2026	Mayand Dhami	Q5 answered; test plan for QR-03 and QR-04 committed.
Free-tier hosting limits checked against the read cycle (R6)	20 November 2026	Isaam Memon	ADR-04 evidence written; R6 closed or escalated.
Build of MUST requirements and P3 baseline	4 December 2026	All six members	MUST requirements pass their acceptance criteria; document updated to v2.0.

10.3 Risks (updated)

Risk	Status since P1	Likelihood / impact	Mitigation and owner
R1: Library availability cannot be read by a program, or changes format (A1)	Open / in progress via Spike S-01	Medium / High	Arshan Khimani runs the spike. If it fails, re-scope via change control to demonstrate against recorded sample data.
R2: Library or OIT objects to automated reading (A2)	Open	Low / High	Lee Baillard contacts the library (Q2). QR-05 caps the request rate at 20 req/min; reading is paused immediately if requested.
R3: NetID SSO cannot be used by a student project (A3)	Open	Medium / Medium	Lee Baillard inquires with OIT (Q3). If refused, Alexander Eng raises a change request to CON-01 to permit email-verified accounts.
R4: Members unavailable around midterms	Active monitoring	Medium / High	Isaam Memon reviews progress weekly; every task has a named backup recorded in its issue. If behind, drop FR-14 then FR-08 via change control.
R5: Alerts arrive too late or sit unseen (A5, QR-01)	Under verification	Medium / Medium	Arshan Khimani measures alert delay with scripted releases. Mayand Dhami revises the threshold based on survey evidence if needed.
R6: Free-tier hosting limits (idle sleep, quotas) stop the 2-minute read cycle (QR-02, CON-03, ADR-04)	New in P2	Medium / High	Isaam Memon checks the chosen provider's idle and quota rules before the build starts. If a limit blocks the cycle, Arshan Khimani raises a change request to QR-02 or moves to another free tier.

10.4 Repository

- **Repository URL:** https://github.com/PaperBagTrick/Study-Room-Sniper
- **Folder layout:** /docs/p2, /docs/p2/diagrams, /spikes/S-01, /docs/elicitation
- **Branching and commits:** main holds only reviewed work merged via pull requests with at least one review. Short-lived feature branches are named type/short-description. Commit messages reference issue numbers (for example, "#12 add FR-04...").
- **Git tag:** Tag p2 points directly to the commit containing the submitted P2 PDF.

10.5 Use of AI tools in P2

Member	Tool	What it was used for	What you checked or changed
Alexander Eng	Claude (Anthropic)	Drafted the initial project concept, text of sections 1–6, requirements, use cases, and diagram scripts for P1.	Checked facts against library policies [R2]–[R4] and verified testability of acceptance criteria.
Alexander Eng	Claude (Anthropic), web search	Looked up UTD library room counts and booking rules.	Opened library pages [R2]–[R4] and cross-verified facts directly.
Alexander Eng	Claude (Anthropic)	Drafted the text of sections 4.2 to 4.3, 5.1, 5.3, 6.1 to 6.6, 7, 8 (set-up only), 9 and 10.2 from the P1 requirements and the team's diagrams.	Checked that requirement IDs, class names and component names match sections 3 and 5 and Figures 5.1, 6.3 and 6.4; spike results and repository evidence were not drafted by Claude.

11 Open Questions (updated; was P1 section 5)

ID	Question	Raised by	Owner and due date	Status or answer	Assumption meanwhile
Q1	Does the library reservation system give programs a supported way to read room availability (feed, API or public page), and on what terms?	Team	McDermott Library staff / Arshan Khimani (27 Oct)	Tested via Spike S-01	Yes: availability can be read without logging in (A1).
Q2	May an unofficial service read availability automatically at up to 20 requests per minute, and tell students about openings?	Team	McDermott Library staff / Lee Baillard (27 Oct)	Pending inquiry	Yes, read-only. We pause reading at once if asked.
Q3	Can a student project use NetID for single sign-on, and what identity data (NetID, email) would it receive?	Team	UTD OIT / Lee Baillard (30 Oct)	Pending inquiry	Yes, and we receive NetID and email address.
Q4	What makes a free slot bookable for a particular student: 7-day horizon, 4-hour max, 1 active reservation per 24 hours [R3]? Should alerts take existing reservations into account?	Team	Library staff / Lee Baillard (30 Oct)	Resolved conceptually	A slot shown free is bookable by any student; alerts ignore the student's individual active bookings.

ID	Question	Raised by	Owner and due date	Status or answer	Assumption meanwhile
Q5	Should a watch stay Active after it alerts, and are a 3-minute alert time and 5-watch limit acceptable?	Team	UTD students (survey) / Mayand Dhami (25 Oct)	Survey administered	Watch stays Active until expired or cancelled; 3 minutes and 5 watches are accepted.
Q6	Which accessibility standard does UTD require of a campus-facing service?	Team	UTD OIT / Lee Baillard (30 Oct)	Pending inquiry	WCAG 2.2 Level AA [R5].

12 Baseline v1.0 Declaration (was P1 section 6)

We declare this document, version 1.0, the baseline for P2. Sections 1 to 9 describe what we will build in P3. Any later change to them is recorded in section 0.2 with its reason.

Team member	Signature or initials	Date
Alexander Eng	Alexander Eng	11/08/2026
Arshan Khimani	Arshan Khimani	11/08/2026
Lee Baillard	Lee Baillard	11/08/2026
Mayand Dhami	Mayand Dhami	11/08/2026
Henry McGahan	Henry McGahan	11/08/2026
Isaam Memon	Isaam Memon	11/08/2026